

BAMM PECHS GOVT. COLLEGE FOR WOMEN

PRELIM EXAMINATION 2026

Time: 2 Hour 40 Minutes

CHEMISTRY PAPER – II

(Marks: 36)

SECTION 'B'
(SHORT-ANSWER QUESTIONS)

Note: Answer any Nine part questions. Select Four part questions from Inorganic-General Chemistry and Five part questions from Organic Chemistry. All part questions carry equal marks.

(INORGANIC-GENERAL CHEMISTRY)

Q2.

- (i) Describe group trend of atomic radius and electronegativity in s block
OR

Describe reasons of the following

- Acidity of hydrogen halides increases from HF to HI
- Electronegativity decreases top to bottom in group. Why?
- Fluorine is the strongest oxidizing agent.
- Multidentate ligands are known as chelating agents.

- (ii) What is flame test? Mention the color of flame of Alkali metals?

- (iii) Why Cu^{+2} is blue but Zn^{+2} is colorless?

- (iv) Describe the preparation and two properties of Nylon and Polyvinyl chloride.

- (v) What is industrial smog and how it is formed?

- (vi) What are outer transition elements? Describe any two of the following properties

- Variable oxidation state
- Colour formation
- Paramagnetic behaviour

- (vii) Write the IUPAC names of the following complexes:

- $\text{Na}_2[\text{Pt}(\text{OH})_4]$
- $[\text{Zn}(\text{NH}_3)_4]^{+2}$

- $[\text{Cu}(\text{en})_2]^{+2}$
- $[\text{Ni}(\text{SCN})_4]^{2-}$

(ORGANIC CHEMISTRY)

- (viii) Define functional group and write the structure of three oxygen containing functional group?

- (ix) Define the following terms

- Catenation
- Isomerism

- Homologous series
- Bucky ball

- (x) Draw the structure of ethene. By a chemical test distinguish between ethene and ethyne

OR

How are amines basic in nature? How can we prepare ethyl amine from the following compounds?

- ethyl iodide
- methyl cyanide
- Ethanamide

2 / 13

- ✓(xi) Bring about the following conversions
- Toluene to ortho-para nitro benzoic acid
 - Benzene to m-nitro toluene

(xii) Define chiral carbon? Give an example to justify your answer?

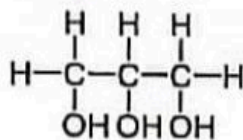
OR

Identify each of the following with one laboratory test:

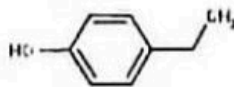
- a. Alcohol *→ Ester test* c. Alkene
b. Phenol *→ Ph. test* d. Aldehyde

- (xiii) Write the IUPAC names of the following organic compounds

- a. $\text{CH}_3\text{-CO-OCH}(\text{CH}_3)_2$
b.



c.



- d. $\text{CH}_2=\text{CH}-\text{C}\equiv\text{C}-\text{CH}=\text{CH}_2$

- (xiv) Write the chemical equation for the following reactions

- a. Reaction of chlorine with acetylene ---
b. Reduction of acetic with LiAlH_4
c. Reaction of benzene with conc HNO_3
d. Hydrolysis of diazonium salt

SECTION 'C'

(DETAILED-ANSWER QUESTIONS)

(Marks:

Note: Answer any Four questions. Select Two question from Inorganic-General Chemistry
Two question from Organic Chemistry. All questions carry equal marks.

(INORGANIC-GENERAL CHEMISTRY)

3. Draw a flow diagram of contact process and describe various steps involved in the synthesis of sulphuric acid?
4. Write the balance equations for the chemical process?
- Sodium burns in excess of air
 - Magnesium is heated with nitrogen gas

- iii. A piece of copper dipped in concentrated sulphuric acid
- iv. Reaction of iron with chlorine
- v. Reaction of conc nitric acid with copper
- vi. Ferric chloride reaction with NaOH
- vii. Chromium reaction with steam
- viii. Phosphorous is put in water

OR

What is bleaching powder? How is it prepared? Give the reaction of bleaching powder with water and hydrochloric acid?

5. Define Troposphere and stratosphere. Describe the chemistry involved due to the presence of oxides of carbon and nitrogen in the troposphere. (Write equations where necessary)

$C_2H_5OH + HCl \rightarrow$
(ORGANIC CHEMISTRY)

6. What are Organo metallic compounds? How is Grignard reagent prepared?

7. Draw the structure of the following organic molecules:

- a. Isopropyl butanoate
- b. Ethyl neo-pentyl ether
- c. Divinylacetylene
- d. p-cresol
- e. Pyrogallol
- f. Neo Pentyl iodide
- g. Benzamide
- h. Benzene-1, 4-dioic acid

OR

Write the equations for the following reactions:

- a. Oxidation of 2°-alcohol with $K_2Cr_2O_7/H_2SO_4$
- b. Reaction of phenol with H_2SO_4 at $20^\circ C$
- c. Reaction of propanone with hydroxyl amine.
- d. Ethanol, in excess, is heated up to $140^\circ C$ in presence of H_2SO_4
- e. Dehydration of ethyl alcohol at $170^\circ C$ in conc. H_2SO_4
- f. Reaction of ethylene glycol with periodic acid.
- g. Reduction of acetone with $LiAlH_4$
- h. Oxidation of 1°-alcohol with PCC.

8. What are lipids? Classify on the basis of structure and give their biological significance.

OR

Write the equation for the nucleophilic addition reaction if formaldehyde treated with

- i. Hydrogen cyanide
- ii. Primary alcohol
- iii. Methyl magnesium bromide
- iv. Ammonia